

Rishikanth Manimeli

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Education

University of Massachusetts Amherst

Expected Graduation: 12/2026

MS in Computer Science | GPA: 3.93/4.0

- **Coursework:** Systems for Deep Learning, Distributed & Operating Systems, Reinforcement Learning, Machine Learning, Database Design, Network Security, Cryptography.

Indian Institute of Technology Patna

07/2018 – 05/2022

Bachelor of Technology in Electrical Engineering

Experience

Samsung R&D

Senior Software Engineer | C++, GTest, Python, Linux

06/2022 – 01/2025

- Optimized MAC Uplink scheduling and core system modules in **C++** to surpass competitive benchmarks for major North American carriers. Resolved CPU bottlenecks via variable access caching, streamlined conditional checks, and protocol-aligned algorithms, increasing LTE throughput by **12-14%** (boosting tested uplink capacity from an advertised 50,000 kbps to ~57,000 kbps per cell for up to 1,200 UEs).
- Developed an automated log analysis application in **Python** to streamline debugging processes, drastically reducing manual log analysis time from **30 minutes** to **5 minutes** per issue for the engineering team.
- Drove architectural improvements to the modem scheduler by contributing **100+ commits** and completing **300+ code reviews**, significantly elevating code quality and system stability.

Projects

Agentic RAG Research Extern | Adobe

01/2026 – 05/2026

- Engineered a multi-document question-answering pipeline utilizing **FlashRAG** and **LLMs** to index 1,000+ research papers, achieving sub-second retrieval latency.
- Formulated a multi-hop reasoning **benchmark** on the Enron email corpus to rigorously evaluate agentic retrieval, circumventing data pollution to ensure valid model assessment.
- Orchestrated an agentic RAG workflow by implementing **tool calling** mechanisms to dynamically retrieve missing context mid-inference, successfully reducing hallucination rates.

In-Memory Key-Value Store | Go

- Engineered a custom Redis-compatible data store in **Go**, leveraging **goroutines** and **TCP networking** to parse the RESP protocol and handle concurrent client connections.
- Designed a leader-follower replication system, guaranteeing full state synchronization and real-time command propagation across distributed nodes.
- Integrated robust data persistence by implementing TTL expiration scheduling and RDB file parsing, enabling seamless recovery from storage failures.

Stock Market Application | Python, AWS

- Architected a fault-tolerant microservices trading platform deployed on **AWS**, implementing **LRU caching** with server-push invalidations and leader-based failover to maintain high availability.
- Applied **Raft** and **Multi-Paxos** consensus algorithms to guarantee totally ordered writes and tolerate node failures via a majority quorum.
- Validated system resilience through extensive load-testing, ensuring strict data integrity, low latency, and correct cache behavior under high concurrent multi-user traffic.

Database Implementation | Java

- Engineered a custom relational Database Management System (DBMS) from scratch in **Java**, implementing core storage components such as a buffer pool manager, on-disk page management, and a heap file engine.
- Constructed a **B+ tree** index structure to optimize query execution, significantly improving database lookup times from $\mathcal{O}(N)$ to $\mathcal{O}(\log N)$, and integrated an iterator-based query executor supporting **Block Nested Loop (BNL) joins**.

Technologies

Programming Languages: C++, Python, Java, Go, Rust, C, TypeScript

Software Development & Tools: AWS, Docker, Git, Linux, PostgreSQL, MongoDB, REST APIs

Frameworks & Libraries: Qt, GTest, FlashRAG

Domains & Methodologies: Large Language Models, Agentic AI, Distributed Systems, Machine Learning, NLP

Leadership & Achievements

Course Grader & Project Mentor (COMPSCI 426): Managed Agile sprints for an 8-student team building an event-ticketing application, evaluating system architecture and functional correctness (Spring 2026).

Samsung Excellence Award: Recognized for exceptional contributions to system capacity improvement and core code optimization (April 2023).

Software Excellency Certification: Earned Samsung's Professional-level certification in algorithmic problem solving (October 2023).